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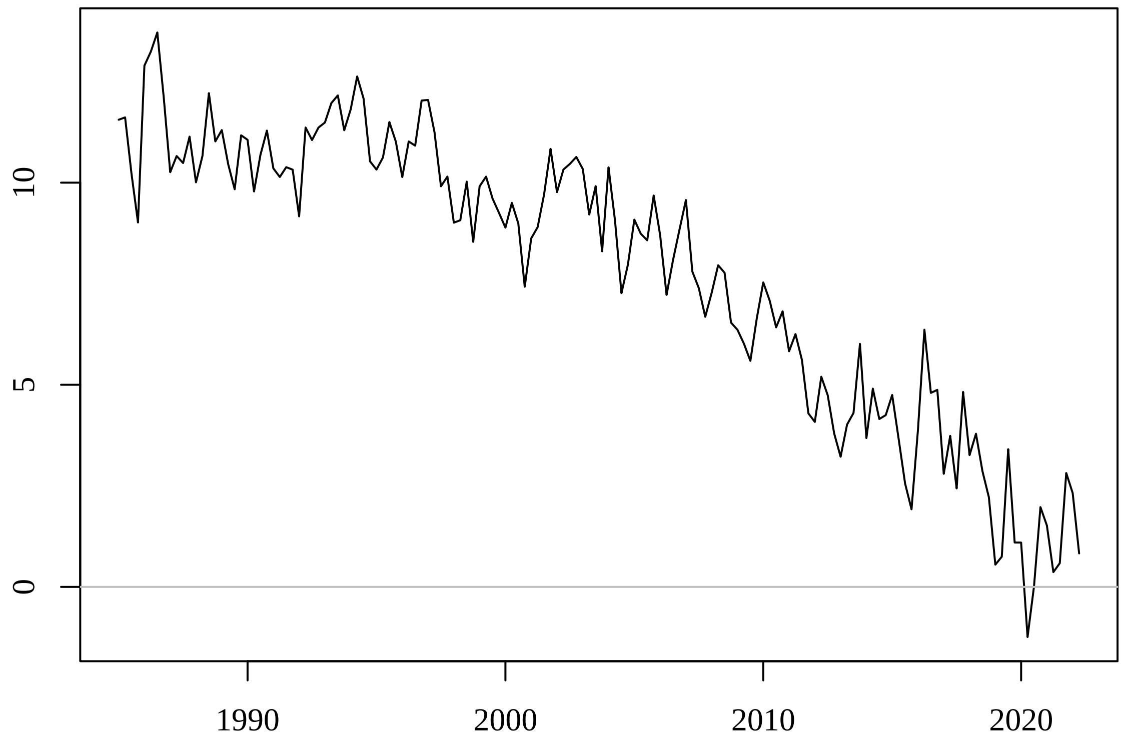
Question 1. (8 marks)

The following is a plot of a quarterly time series from 1985q1 to 2022q2, a total of 150 observations.



(a) **(2 marks)** Does this time series plot suggest a **trend** is present? Answer "yes", "no" or "can't tell", with a one sentence explanation.

(1 mark) Yes.

(1 mark) The plot suggests a trend because the time series shows an overall downwards movement during this time.

(b) (2 marks) Does the time series plot suggest a **trend break** is present? Answer "yes", "no" or "can't tell", with a one sentence explanation.

(1 mark) Yes.

(1 mark) The slope of the downwards trend appears to become steeper around the middle of the sample.

(c) (2 marks) Does the time series plot suggest **non-stationarity** is present? Answer "yes", "no" or "can't tell", with a one sentence explanation.

(1 mark) Yes.

(1 mark) The presence of the trend suggests the mean of the time series is decreasing over time, which implies non-stationarity.

(d) (2 marks) Does the time series plot suggest **non-invertibility** is present? Answer "yes", "no" or "can't tell", with a one sentence explanation.

(1 mark) Can't tell.

(1 mark) We cannot tell whether a moving average component is present, nor what inverse root(s) it may have.

Question 2. (15 marks)

An augmented Dickey-Fuller test regression with trend was calculated for the time series in the plot in Question 1 using the `urca` command `ur.df`:

```
ur.df(Y, type="trend", lags=1)
```

The following test regression was obtained.

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	3.860	0.895	4.314	0.000
z.lag.1	-0.289	0.066	-4.413	0.000
tt	-0.022	0.005	-4.264	0.000
z.diff.lag	-0.085	0.083	-1.028	0.306

(a) (2 marks) Write out the estimated test regression in equation form.

$$\Delta\widehat{Y}_t = 3.860 - 0.289Y_{t-1} - 0.022t - 0.085\Delta Y_{t-1}$$

(b) (1 mark) What is the value of the augmented Dickey-Fuller t test statistic?

$t = -4.413$

(c) (2 marks) What are the null and alternative hypotheses of the Dickey-Fuller test?

$H_0 : Y_t$ has a unit root, or $H_0 : \varphi = 0$

$H_1 : Y_t$ does not have a unit root, or $H_1 : \varphi < 0$

If the latter, define $\Delta Y_t = \beta_0 + \varphi Y_{t-1} + \beta_1 t + \psi_1 \Delta Y_{t-1} + U_t$.

The `urca` command `qunitroot(q, trend="ct")` is used to compute the critical values for the augmented Dickey-Fuller test for `q = 0.05, 0.01, 0.005, 0.001`. The output is shown here:

```
                       5%     1%   0.5%   0.1%
ADF critical value: -3.41 -3.958 -4.164 -4.596
```

(d) **(3 marks)** Carry out the augmented Dickey-Fuller test using the statistic in part (b) using the 5% level of significance. State the test decision and conclusion.

(2 marks) $t = -4.413$ is less than the 5% critical value -3.41, therefore reject H_0.

(1 mark) The conclusion is that Y_t does not have a unit root.

(e) **(3 marks)** Suppose we decide there is a break in the trend line occuring in quarter 1 of 2003, which is observation $t = 80$, such that the trend line remains continuous at that point. Write out the form of the test equation that would be estimated to include this trend break. Clearly define any additional notation you need.

(2 marks) The test equation has the form

$$\Delta Y_t = \beta_0 + \varphi Y_{t-1} + \beta_1 t + \beta_2 TB_t + \psi_1 \Delta Y_{t-1} + U_t$$

where

(1 mark) $TB_t = t - 80$ for $t \geq 80$, and $TB_t = 0$ otherwise.

(f) **(5 marks)** Outline the steps involved in computing an appropriate 5% critical value for the test in part (e).

(4 marks) Repeat simulation steps (say) 10,000 times:

- generate $U_t \sim$ i.i.d.$N(0, 1)$ for $t = 1, \ldots, 150$
- compute $Y_t = Y_{t-1} + U_t$
- estimate the test equation from part (e), excluding ΔY_{t-1}, and obtain the t statistic on Y_{t-1}

(1 mark) For the critical value, compute the 5% percentile from the 10,000 replications of the t statistic.

If you run out of space for Question 2, use the extra pages at the end of the exam paper. Please clearly indicate there the question number/part you are answering.

☐ **Please tick this box if you used extra pages for any part of Question 2.**

Question 3. (23 marks)

For a stationary time series with $n = 200$ observations a range of ARMA(p,q) were fitted to attempt to find a suitable forecasting model. The following table shows the values of the AICc and p-values of the Ljung-Box residual autocorrelation test for every combination of $p = 0, 1, 2, 3, 4$ and $q = 0, 1, 2$.

p	q	AICc	LBp
0	0	1.070	0.000
1	0	1.016	0.048
2	0	1.011	0.148
3	0	1.005	0.872
4	0	1.008	0.311
0	1	1.000	0.835
1	1	1.003	0.800
2	1	1.007	0.647
3	1	1.008	0.881
4	1	1.009	0.287
0	2	1.003	0.841
1	2	1.007	0.796
2	2	1.010	0.764
3	2	1.011	0.909
4	2	1.013	0.292

(a) **(4 marks)** Based on the Ljung-Box p values (`LBp`), which model(s) may be considered to be **mis**specified? Justify your conclusions.

(2 marks) The ARMA(0,0) and ARMA(1,1) models have LB $p < 0.05$, hence the null hypothesis of no residual autocorrelation is rejected for each of these models.

(2 marks) Residual autocorrelation implies misspecification of the model for the conditional mean, since prediction errors be definition cnanot be autocorrelated.

(b) **(3 marks)** According to the AICc alone, what is the best model? Is there evidence this model is misspecified? Explain.

(1 mark) MA(1) or ARMA(0,1)

(1 mark) has the smallest AICc.

(1 mark) This model has LB $p > 0.05$, suggesting no evidence of model misspecification.

(c) **(3 marks)** According to the AICc alone, what is the best AR(p) model? Is there evidence this model is misspecified? Explain.

(1 mark) AR(3)

(1 mark) has the smallest AICc.

(1 mark) This model has LB $p > 0.05$, suggesting no evidence of model misspecification.

The coefficient estimates of the ARMA(0,1) and ARMA(3,0) models are as follows:

```
               ma1 intercept
ARMA(0,1): -0.491     3.014

               ar1    ar2    ar3 intercept
ARMA(3,0): -0.476 -0.234 -0.166     3.014
```

(d) (4 marks) Write out both models in equation form.

(2 marks) $\text{ARMA}(0,1): Y_t = 3.014 + \widehat{Z}_t$

$$\widehat{Z}_t = -0.491 U_{t-1}$$

(2 marks) $\text{ARMA}(3,0): Y_t = 3.014 + \widehat{Z}_t$

$$\widehat{Z}_t = -0.476 Z_{t-1} - 0.234 Z_{t-2} - 0.166 Z_{t-3}$$

Forecasts from the ARMA(0,1) and ARMA(3,0) models for $h = 1, 2, \ldots, 8$ steps ahead are tabulated here:

h	ARMA(0,1)	ARMA(3,0)
1	2.993	3.065
2	3.014	3.052
3	3.014	2.952
4	3.014	3.025
5	3.014	3.016
6	3.014	3.020
7	3.014	3.008
8	3.014	3.014

(e) **(5 marks)** The two models have quite different functional forms and coefficient estimates, but quite similar forecasts. Give an explanation for this.

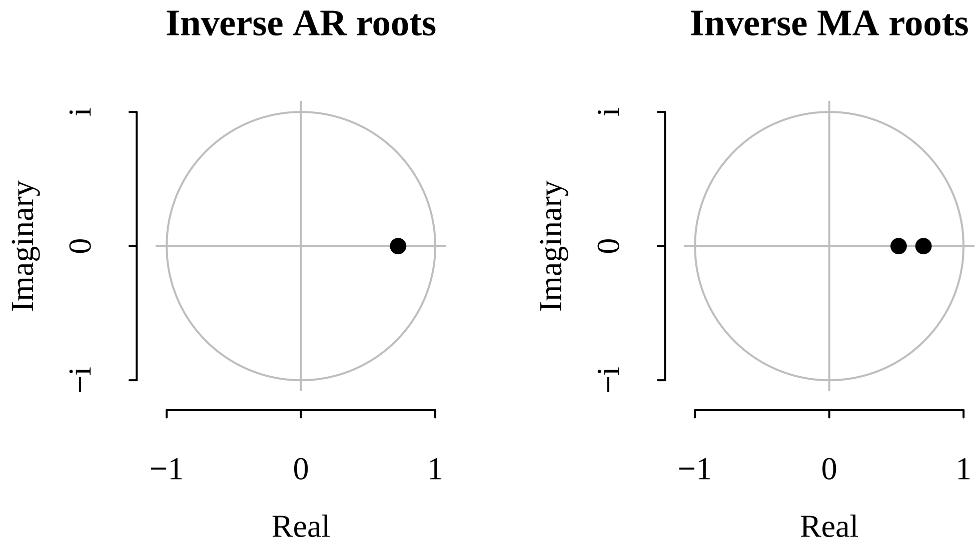
(1 marks) The MA(1) is invertible
(1 marks) because the MA(1) coefficient is less than one in magnitude.

(1 mark) Therefore the MA(1) model could be inverted and be approximated by AR(p) model.

(1 mark) The AR(3) model is selected as the best AR(p) approximation to the best model.

(1 mark) The forecast are quite similar because the AR(3) approximation is quite good for the MA(1) model.

Referring to the table of AICc values, the ARMA(1,2) model may be another candidate model to try. The estimation of this model produces the following inverse roots.




```
Inverse MA roots: 0.722 0.517
Inverse AR root:  0.723
```

(f) (7 marks) Give an explanation of what information these two plots suggest about the specification of the ARMA(1,2) model.

(2 marks) The MA inverse roots are inside the unit circle, implying invertibility.

(2 mark) The AR inverse root is inside the unit circle, implying stationarity.

(2 marks) The larger MA inverse root is almost exactly equal to the AR inverse root. This implies the presence of a common factor on either side of the ARMA model,

(1 mark) i.e. that the AR(1) and MA(2) components can be cancelled, reducing to the MA(1) model which was best according to the AICc.

If you run out of space for Question 3, use the extra pages at the end of the exam paper.

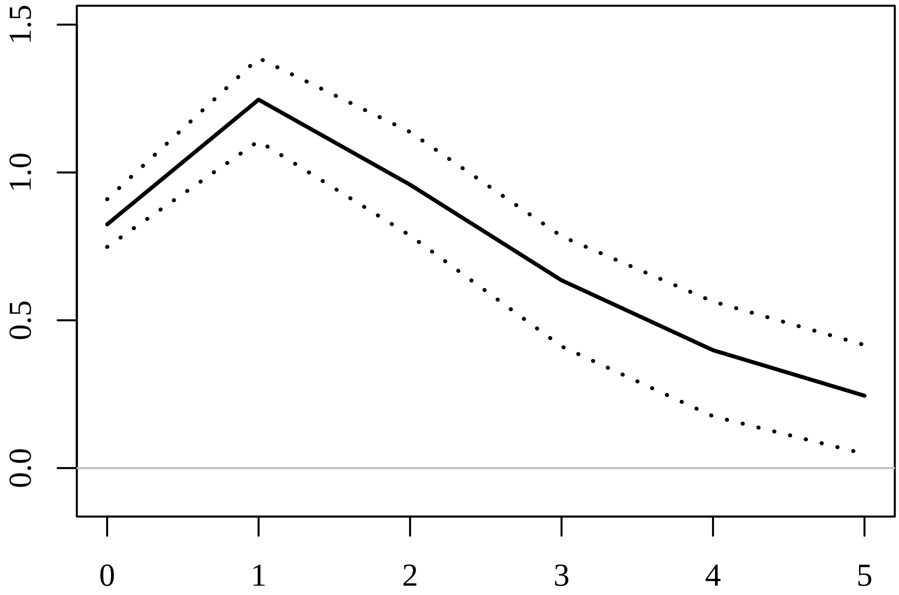
☐ **Please tick this box if you used extra pages for any part of Question 3.**

Question 4. (10 marks)

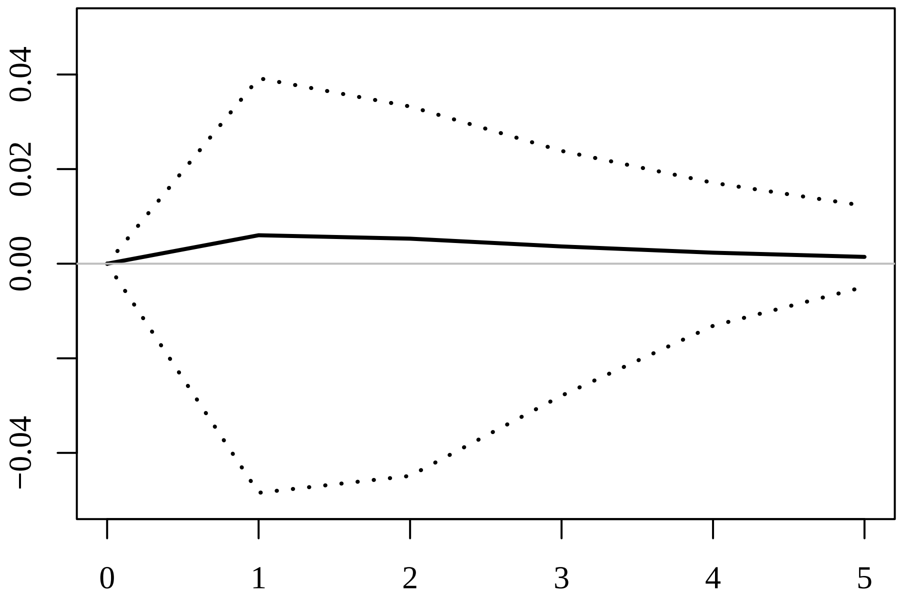
We have $n = 300$ observations on two time series, $Y_{1,t}$ and $Y_{2,t}$, to use as estimation sample, with the following $h = 5$ observations retained for forecast evaluation purposes. Following specification searches, bivariate VAR(1) model (`VAR`).

Using the VAR(1) model for $Y_{1,t}$ and $Y_{2,t}$, *orthogonalised* impulse response functions were calculated and plotted below, along with 95% confidence intervals (in dashed lines).

Response of $E(Y_{2,n+h}|\mathscr{Y}_n)$ to impulse to $Y_{1,n}$



Response of $E(Y_{1,n+h}|\mathscr{Y}_n)$ to impulse to $Y_{2,n}$



(a) (10 marks) Provide full interpretations of these two impulse response functions.

Response of $E(Y_{2,n+h}|\mathscr{Y}_n)$ to impulse to $Y_{1,n}$

(2 marks) The magnitude of the impulse to $Y_{1,n}$ is not shown, but the magnitude of the contemporaneous impulse to $Y_{2,n}$ is about 0.8, shown at $h = 0$. This is computed from the Choleski factorisation of the conditional variance matrix of $Y_{1,t}$ and $Y_{2,t}$ based on the VAR(1) model residuals.

(1 mark) The subsequent impulse responses for $h > 0$ show the extent of revisions to the forecasts of $Y_{2,n+h}$ because of the impulse at time n.

(2 marks) The impulse responses are all positive and significant because the 95% confidence intervals do not include 0.

(1 mark) The impulse responses decay towards 0 as h reaches 5.

Response of $E(Y_{1,n+h}|\mathscr{Y}_n)$ to impulse to $Y_{2,n}$

(2 marks) The contemporaneous impulse to $Y_{1,n}$ is zero, which is imposed by the ordering of variable in the VAR, i.e. this is not an estimate.

(2 marks) The impulse responses for $h > 0$ are all positive but insignificant because the 95% confidence intervals all include 0.

If you run out of space for Question 4, use the extra pages at the end of the exam paper. Please clearly indicate there the question number/part you are answering.

☐ **Please tick this box if you used extra pages for any part of Question 4.**

Question 5. (14 marks)

Consider a model for a quarterly time series Y_t:

$$Y_t = \beta_0 + \beta_1 Q_{1,t} + \beta_2 Q_{2,t} + \beta_3 Q_{3,t} + Z_t$$
$$Z_t = 0.3 Z_{t-1} + 0.4 Z_{t-2} + U_t + 0.5 U_{t-1},$$

where $Q_{1,t}$, $Q_{2,t}$ and $Q_{3,t}$ are seasonal dummy variables taking the value 1 in quarters 1, 2 and 3 respectively, and the value 0 otherwise.

(a) **(2 marks)** Express the equation for Z_t using lag operators and polynomials.

(2 marks) $(1 - 0.3L - 0.4L^2)Z_t = (1 + 0.5L)U_t$

(b) **(4 marks)** Is it possible that the model for Z_t is stationary? Explain why or why not, and why the answer needs to be qualified by "is it possible".

(1 marks) The autoregressive lag polynomial can be factored as

$$(1 - 0.8L)(1 + 0.5L).$$

(1 mark) This implies inverse roots of 0.8 and -0.5, both inside the unit circle.

(1 mark) Because of this, it is possible that the AR model is stationary...

(1 mark) ...provided appropriate assumptions about the mean and variance of the initial values are made to ensure the mean and variance of Z_t are constant for all t. (No need to try to derive these.)

(c) **(2 marks)** Is the model for Z_t is invertible? Explain why or why not.

(1 mark) The MA lag polynomial $(1 + 0.5L)$ has inverse root 0.5, which is inside the unit circle.

(1 mark) Therefore it is invertible.

(d) **(2 marks)** Does the model for Z_t contain a common factor? Explain why or why not.

(2 marks) The representation

$$(1 - 0.8L)(1 + 0.5L)Z_t = (1 + 0.5L)U_t$$

reveals a common factors, i.e. the model could be cancelled down to AR(1).

(e) **(4 marks)** Is it possible that the entire model for Y_t (i.e. both equations considered together) is stationary? If necessary, explain any special case(s) that affect your general answer.

(2 marks) In general Y_t will be non-stationary because the seasonal dummies have the effect of producing a different mean $E(Y_t)$ depending on the season.

(2 marks) In the special case of $\beta_1 = \beta_2 = \beta_3 = 0$ then Y_t can be stationary because then it has constant mean β_0 and otherwise inherits the stationarity properties of Z_t in part (b).

If you run out of space for Question 5, use the extra pages at the end of the exam paper. Please clearly indicate there the question number/part you are answering.

☐ **Please tick this box if you used extra pages for any part of Question 5.**

Question 6. (16 marks)

Let Y_t be a time series that satisfies the following AR(1)-ARCH(1) model

$$Y_t = \beta_0 + Z_t$$
$$Z_t = \phi_1 Z_{t-1} + U_t$$
$$E(\,U_t \,|\, \mathscr{Y}_{t-1}\,) = 0$$
$$E(\,U_t^2 \,|\, \mathscr{Y}_{t-1}\,) = \omega + \alpha_1 U_{t-1}^2.$$

(a) **(2 marks)** Is it valid to consider U_t as the *one-step-ahead prediction error* in this model when it has the ARCH(1) structure?

(2 marks) It is valid because $E(\,U_t \,|\, \mathscr{Y}_{t-1}\,) = 0$ implies U_t is not 1-step-ahead predictable. The conditional variance structure does not affect this.

(b) **(2 marks)** Why is it required to impose $\omega > 0$ and $\alpha_1 > 0$ in this model?

(2 marks) It is necessary for the model to imply positive values for $E(\,U_t^2 \,|\, \mathscr{Y}_{t-1}\,)$, and the imposition of $\omega > 0$ and $\alpha_1 > 0$ (along with the squared in U_{t-1}^2) guarantees this.

(c) (6 marks) What parameter restriction is required for U_t to be stationary? Under this restriction, derive the *unconditional* mean and variance of U_t.

(1 mark) The stationarity condition for U_t is $\alpha_1 < 1$ as will become evident.

(1 mark) The unconditional mean follows from the LIE:

$$E(U_t) \stackrel{\text{LIE}}{=} E(\, E(\, U_t \,|\, \mathscr{Y}_{t-1}\,)\,) = 0$$

(2 marks) The unconditional variance follows from

$$\begin{aligned} \operatorname{var}(U_t) &= E(U_t^2) \stackrel{\text{LIE}}{=} E(\, E(\, U_t^2 \,|\, \mathscr{Y}_{t-1}\,)\,) = E(\, \omega + \alpha_1 U_{t-1}^2\,) \\ &= \omega + \alpha_1 E(U_{t-1}^2) \end{aligned}$$

(1 mark) Stationarity requires $E(U_t^2) = E(U_{t-1}^2)$ for all t, so that

$$E(U_t^2) = \omega + \alpha_1 E(U_t^2)$$

(1 mark) which solves to give

$$E(U_t^2) = \frac{\omega}{1-\alpha_1},$$

which requires $\alpha_1 < 1$ to be valid.

(d) **(4 marks)** Give expressions for the conditional mean and conditional variance of Y_t, i.e. $E(Y_t \mid \mathscr{Y}_{t-1})$ and $\text{var}(Y_t \mid \mathscr{Y}_{t-1})$, *as functions of lags of* Y_t.

(2 marks) Substitution of $Z_t = Y_t - \beta_0$ gives

$$Y_t = \beta_0 + \phi_1(Y_{t-1} - \beta_0) + U_t$$

and hence

$$E(Y_t|\mathscr{Y}_{t-1}) = \beta_0 + \phi_1(Y_{t-1} - \beta_0)$$

since $E(U_t|\mathscr{Y}_{t-1}) = 0$.

It also implies that

$$\text{var}(Y_t|\mathscr{Y}_{t-1}) = \text{var}(U_t|\mathscr{Y}_{t-1}).$$

(2 marks) Substitution for U_{t-1}, Z_{t-1} and Z_{t-2} gives

$$U_{t-1} = (Y_{t-1} - \beta_0) - \phi_1(Y_{t-2} - \beta_0)$$

so that

$$\begin{aligned}\text{var}(Y_t|\mathscr{Y}_{t-1}) = \text{var}(U_t|\mathscr{Y}_{t-1}) &= \omega + \alpha_1 U_{t-1}^2 \\ &= \omega + \alpha_1((Y_{t-1} - \beta_0) - \phi_1(Y_{t-2} - \beta_0))^2\end{aligned}$$

(e) **(2 marks)** How would such expressions for $E(Y_t \mid \mathscr{Y}_{t-1})$ and $\text{var}(Y_t \mid \mathscr{Y}_{t-1})$ be useful for constructing a normal 95% prediction interval for the one-step-ahead forecast $E(Y_{n+1} \mid \mathscr{Y}_n)$?

(2 marks) The normal 95% prediction interval formula is

$$\widehat{E}(Y_{n+1}|\mathscr{Y}_n) \pm 1.96 \times \sqrt{\widehat{\text{var}}(Y_{n+1}|\mathscr{Y}_n)}$$

Expressions for the conditional mean and variance therefore provide the formulae that are used to compute the prediction interval once the parameters are estimated.

(f) **(2 marks)** Referring to your answer to (d), which lags of Y_n are used in computing the prediction interval in the AR(1)-ARCH(1) model in this question?

(2 marks) Having estimated the model parameters, we would have

$$\widehat{E}(Y_{n+1}|\mathscr{Y}_n) = \widehat{\beta}_0 + \widehat{\phi}_1(Y_n - \widehat{\beta}_0)$$
$$\widehat{\mathrm{var}}(Y_{n+1}|\mathscr{Y}_n) = \widehat{\omega} + \widehat{\alpha}_1((Y_n - \widehat{\beta}_0) - \widehat{\phi}_1(Y_{n-1} - \widehat{\beta}_0))^2$$

The prediction interval for Y_{n+1} therefore depends on Y_n and Y_{n-1}.

(g) **(2 marks)** Referring to your answer to (d), how does value of the one-step-ahead forecast for Y_{n+1} relate to the value of the ARCH(1) coefficient (α_1)?

(2 marks) $\widehat{E}(Y_{n+1}|\mathscr{Y}_n) = \widehat{\beta}_0 + \widehat{\phi}_1(Y_n - \widehat{\beta}_0)$

The point forecast does not depend on the ARCH(1) coefficient at all.

(h) **(2 marks)** Referring to your answer to (d), how does the width of the one-step-ahead prediction interval for Y_{n+1} relate to the value of the ARCH(1) coefficient (α_1)?

(2 marks) The width of the prediction interval is

$$2 \times 1.96 \times \sqrt{\widehat{\mathrm{var}}(Y_{n+1}|\mathscr{Y}_n)}$$

and the expression in (d) for the conditional variance shows that it is increasing in the value of the ARCH(1) coefficient. i.e. the width of the prediction interval has a positive relationship with the ARCH(1) coefficient, all else equal.

Extra page. Clearly indicate each question number/part you are answering.

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